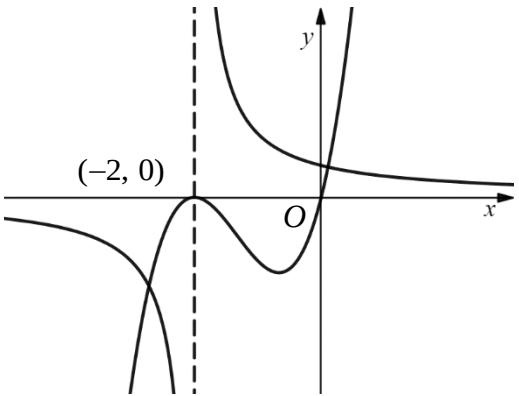
1a	$y = 5x^3 + \frac{3}{x^2} - 7x = 5x^3 + 3x^{-2} - 7x$ $\left(\frac{dy}{dx} =\right) \ 15x^2 - 6x^{-3} - 7$	M1A1A1
		(3)
b	$\left(\frac{d^2y}{dx^2} =\right) \ 30x + 18x^{-4}$	M1A1
		(2)
		(5 marks)

2a	$\frac{1}{8}x$	B1
		(1)
b	$\frac{1}{256}x^{\frac{3}{2}}$	B1
		(1)
c	$\left(\frac{1}{2}\left(\frac{1}{64}x^2 \times \frac{16}{\sqrt{x}}\right)\right)^{-\frac{4}{3}} = \left(\frac{1}{8}x^{\frac{3}{2}}\right)^{-\frac{4}{3}} = 16x^{-2}$	M1A1
		(2)
		(4 marks)

3a	$\frac{8-\sqrt{15}}{2\sqrt{3}+\sqrt{5}}\times\frac{2\sqrt{3}-\sqrt{5}}{2\sqrt{3}-\sqrt{5}}=\frac{16\sqrt{3}-8\sqrt{5}-2\sqrt{45}+\sqrt{75}}{12-5}$	M1
	e.g. $\frac{21\sqrt{3}-14\sqrt{5}}{7}$	dM1
	$3\sqrt{3}-2\sqrt{5}$	A1
		(3)
b	$\left(x+5\sqrt{3}\right)\sqrt{5}=40-2x\sqrt{3}\Rightarrow x\sqrt{5}+2x\sqrt{3}=40-5\sqrt{15}$	M1
	$\left(x=\right)\frac{40-5\sqrt{15}}{2\sqrt{3}+\sqrt{5}}$	A1
	$\left(x=\right)15\sqrt{3}-10\sqrt{5}$	A1ft
		(3)
		(6 marks)

4a	$x = -2$	B1
		(1)
b	$x^3 + 4x^2 + 4x = x(x+2)^2$	M1A1
		(2)
c		B1B1B1
		(3)
d	2 as the graphs intersect (each other) twice (since $(x+2)(x^3+4x^2+4x)=1$ is the same as $x^3+4x^2+4x=\frac{1}{x+2}$)	B1
		(1)
		(7 marks)

5a	e.g. $(\cos\theta =)\ \dfrac{5^2+(1+x)^2-x^2}{2\times5(1+x)}\ \left(=\dfrac{25+1+2x+x^2-x^2}{2\times5(1+x)}\right)$ or e.g. $x^2=5^2+(1+x)^2-2\times5(1+x)\cos\theta$	M1
	$\cos\theta=\dfrac{13+x}{5+5x}$ *	A1*
		(2)
b	$\theta=$ awrt $42°$ (42.470747….)	B1

	Attempts to find *AB*, *AD* or *AC*: e.g. $\dfrac{AB}{\sin"42"}=\dfrac{5}{\sin30}\Rightarrow AB=...$ or e.g. $\angle ABC=180-30-"42.5"="107.6"$ $\dfrac{\sin DBC}{1+2\sqrt{3}}=\dfrac{\sin"42.5"}{2\sqrt{3}}\Rightarrow\angle DBC="60.5",\ \angle ABD="107.6"-"60.5"="47.1"$ $\dfrac{AD}{\sin("47.1")}=\dfrac{2\sqrt{3}}{\sin30}\Rightarrow AD=...$ or e.g. $\dfrac{AC}{\sin"108"}=\dfrac{5}{\sin30}\Rightarrow AC=...$	M1

	$AB=6.75...$ or $AD=5.07...$ or $AC=9.53...$	A1
	Area $=\dfrac{1}{2}\times5\times"6.75"\times\sin(180-30-"42.5")$ or $=\dfrac{1}{2}\times5\times"9.54"\times\sin("42.5")$	dM1
	$=$ awrt 16.1 (m^2)	A1
		(5)
		(7 marks)

6a	$4(p-2x)=\dfrac{12+15p}{x+p}$ $8x^2+4px+12+15p-4p^2 \quad (=0)$	M1
	e.g. $a=8,\ b=4p,\ c=12+15p-4p^2$ $(4p)^2-4\times8\times(12+15p-4p^2) \quad (>0)$	dM1
	$3p^2-10p-8>0$ *	A1*
		(3)
b	e.g. $(p-4)(3p+2)\ (=0)\Rightarrow 4,\ -\dfrac{2}{3}$	M1
	$p<-\dfrac{2}{3}$ or $p>4$	M1A1
		(3)
		(6 marks)

7ai	$f'(4)=\dfrac{4(4)^2+10-7(4)^{\frac{1}{2}}}{4(4)^{\frac{1}{2}}}=\dfrac{15}{2}$	B1
ii	$"\dfrac{15}{2}"\rightarrow-\dfrac{2}{15}$	M1
	$y+1="-\dfrac{2}{15}"(x-4)$	M1
	$2x+15y+7=0$	A1
		(4)
b	$\dfrac{4x^2+10-7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}=\pm...x^{\frac{3}{2}}\pm...x^{-\frac{1}{2}}\pm...$	M1
	Two of the terms of $x^{\frac{3}{2}}+\dfrac{5}{2}x^{-\frac{1}{2}}-\dfrac{7}{4}$	A1
	$\int\left(x^{\frac{3}{2}}+\dfrac{5}{2}x^{-\frac{1}{2}}-\dfrac{7}{4}\right)\mathrm{d}x=\dfrac{2}{5}x^{\frac{5}{2}}+5x^{\frac{1}{2}}-\dfrac{7}{4}x\ \ (+c)$	dM1A1ft
	$\dfrac{2}{5}(4)^{\frac{5}{2}}+5(4)^{\frac{1}{2}}-\dfrac{7}{4}(4)+c=-1\Rightarrow c=...$	ddM1
	$\left(f(x)=\right)\ \dfrac{2}{5}x^{\frac{5}{2}}+5x^{\frac{1}{2}}-\dfrac{7}{4}x-\dfrac{84}{5}$	A1
		(6)
		(10 marks)

8a	$x\left(x^3-\frac{7}{2}x-5\right)=\frac{15}{2}-5x$ $\Rightarrow 2x^4-7x^2-15=0$ *	M1 A1*
		(2)
b	$(2x^2+3)(x^2-5)=0\Rightarrow\left(x^2=\right)\ldots$	M1
	$\left(x=\right)\ \ (\pm)\sqrt{5}$	B1
	$y=\frac{15}{2\times"\pm\sqrt{5}"}-5=\ldots$ or $y=\left("\pm\sqrt{5}"\right)^3-\frac{7}{2}\left("\pm\sqrt{5}"\right)-5$	M1
	$\left(-\sqrt{5},-\frac{3}{2}\sqrt{5}-5\right)\quad\left(\sqrt{5},\frac{3}{2}\sqrt{5}-5\right)$ $PQ=\sqrt{("2\sqrt{5}")^2+("3\sqrt{5}")^2}=\ldots$	dM1
	$PQ=\sqrt{65}$	A1cao
		(5)
		(7 marks)

9a	$0.72 = OP \times 0.6 \Rightarrow OP = 1.2$ **or** $\frac{0.72}{0.6} = 1.2$ **or** $0.72 = 1.2 \times 0.6 \Rightarrow OP = 1.2$ *	B1*
		(1)
b	$\frac{1}{2} \times (x+1.2)^2 \times 0.6$	B1
	$\frac{1}{2} \times (x+1.2)^2 \times 0.6 - \frac{1}{2} \times 1.2^2 \times 0.6 = 90 \Rightarrow 5x^2 + 12x - 1500 = 0$ *	M1A1*
		(3)
c	$x =$ awrt 16.2 $\left(= \frac{-6 + 4\sqrt{471}}{5} \right)$	B1
	Perimeter $= 0.72 + "16.2" + "16.2" + 0.6 \times ("16.2" + 1.2) =$ awrt 43 / awrt 44 m	M1A1
		(3)
		(7 marks)

10ai	$(n=)\ 3$	B1
ii	1080	B1
		(2)
b	$(1620, -3)$	B1B1
		(2)
c	e.g. $k = \frac{1}{2}\left(\frac{12}{5} - \frac{3}{5}\right) = \frac{9}{10}$	M1A1
		(2)
		(6 marks)

11a	$2(x \pm ...)^2$ $...(x \pm 3)^2 ...$ $2(x-3)^2 - 4$	B1 M1 A1
		(3)
b	$(3, -4)$	B1ft
		(1)
c	$m = \frac{28 - -4}{-1-3}$ $(= -8)$ $y - 28 = -8(x+1)$ $y = -8x + 20$	M1 dM1 A1
		(3)
d	$y \leqslant "-8x+20"$ and $y \geqslant 2x^2 - 12x + 14$ (or $y \geqslant "2(x-3)^2 - 4"$) $y \leqslant -8x + 20$ $\quad y \geqslant 2x^2 - 12x + 14$ $\quad y \geqslant 0,\ x \geqslant 0$	B1ftB1ft B1cso
		(3)
		(10 marks)